Simulation Assumption Limitation Module (SALM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Governed Space

Simulation Assumption Limitations

Canonical Definition

Simulation Assumption Limitation Module (SALM) defines the governance framework for identifying, defining, bounding, classifying, propagating, disclosing, reassessing, and preserving the limitations created by Simulation Assumptions throughout the simulation lifecycle.

It establishes the methodological conditions under which every material Simulation Assumption remains connected to the boundaries within which it may legitimately operate, the conditions under which it may cease to remain applicable, the simulation elements and outputs affected by those limitations, and the consequences those limitations create for interpretation and downstream use.

SALM does not determine whether a Simulation Assumption is universally valid.

It governs where an assumption stops being safely or legitimately applicable inside the simulation and what consequences follow when that boundary is approached, crossed, changed, or remains unknown.

Operational Role

SALM serves as the fifth and final internal governance space of the Simulation Assumption Standard (SAS).

The preceding modules establish:

SAIM — Which assumptions exist and which are materially relevant?

SABM — On what basis is each assumption being used?

SADM — What depends upon each assumption?

SAUM — What remains uncertain about each assumption and where may that uncertainty propagate?

SALM asks:

Within which conditions may the assumption legitimately operate, where does its applicability end, and what limitations does it impose upon the simulation?

The governed progression is:

Governed Simulation Assumption → Limitation Identification → Applicability Boundary → Limiting Condition → Dependency Connection → Limitation Propagation → Consequence Assessment → Disclosure → Reassessment → Limitation Record

SALM closes the internal governance lifecycle of the Simulation Assumption Standard.

Module Operational Space

SALM governs: assumption limitation identification, assumption applicability boundaries, limiting conditions, applicability conditions, environmental limitations, temporal limitations, spatial limitations, population limitations, behavioral limitations, operational limitations, parameter limitations, model-context limitations, scenario limitations, evidence-related limitations, uncertainty-derived limitations, dependency-derived limitations, conditional applicability, partial applicability, non-applicability, limitation thresholds, limitation proximity, limitation breach, limitation propagation, limitation interaction, compound limitations, limitation materiality, limitation consequences, output limitations, interpretation limitations, limitation disclosure, limitation reassessment, limitation change, limitation version binding, historical limitation preservation, limitation gaps, and complete assumption-limitation traceability.

Module Function

Simulation Assumptions allow a simulation to operate where complete representation of reality is impossible, unnecessary, unavailable, or methodologically inappropriate.

But every assumption creates a boundary.

An assumption may be reasonable:

within one environment

but not another.

It may apply:

within one parameter range

but fail outside it.

It may represent:

one population

but not another.

It may remain useful:

during one period

but become obsolete after material change.

SALM prevents these boundaries from disappearing once an assumption has been incorporated into a model.

It transforms:

Simulation Assumption

into:

Simulation Assumption + Applicability Boundary + Limiting Conditions + Affected Dependencies + Consequences Simulation Assumption Limitation

A Simulation Assumption Limitation is a governed condition that restricts where, when, how, or to what extent a Simulation Assumption may legitimately represent the condition for which it is used.

A limitation may constrain: applicability, parameter range, environment, population, time, scenario, operating state, interpretation, simulation output, or downstream conclusions.

Assumption Applicability Boundary

Every material assumption should have an identifiable applicability boundary where reasonably possible.

The boundary answers:

Under which conditions is this assumption intended to apply?

and:

Under which conditions should it no longer be treated as applicable?

Conceptually:

Applicable Conditions → Boundary → Non-Applicable Conditions Applicability

Context

Assumption applicability should be understood relative to:

Simulation Object + Purpose + Scope + Reality Representation + Model + Environment + Scenario

The same assumption may therefore be legitimate in one simulation context and inappropriate in another.

Limiting Condition

A Limiting Condition is a condition that narrows or terminates the legitimate applicability of an assumption.

Examples may include: temperature above a defined range, velocity outside tested conditions, different population characteristics, changed environmental behavior, altered system configuration, different operating mode, unavailable supporting evidence, changed temporal context, or newly discovered dependency.

Conditional Applicability

Some assumptions remain legitimate only while specified conditions remain true.

Conceptually:

Assumption Applicable IF Conditions A + B + C remain satisfied

If a material condition changes, the assumption should enter reassessment.

Partial Applicability

An assumption may remain applicable to only part of the Simulation Object, scenario set, operating envelope, or population.

SALM requires that partial applicability remain visible rather than being represented as universal applicability. Non-Applicability

A Simulation Assumption becomes non-applicable when the conditions required for its legitimate use are no longer satisfied.

Non-applicability should not automatically be interpreted as failure of the entire simulation.

It establishes that the affected simulation pathway can no longer legitimately rely upon that assumption without reassessment, replacement, or explicit restriction.

Environmental Limitation

An assumption may apply only within specified environmental conditions.

Examples include: temperature, humidity, radiation, terrain, atmospheric conditions, lighting, network conditions, infrastructure availability, or another environmental state.

Temporal Limitation

An assumption may be valid only for a defined period or temporal context.

Examples include: market behavior derived from a historical period, component degradation assumptions, demographic conditions, seasonal behavior, or technology-performance assumptions.

A materially changing temporal context may therefore create a limitation.

Spatial Limitation

An assumption may apply only within defined geographic, physical, or spatial conditions.

An assumption derived from one region should not automatically be treated as applicable globally.

Population Limitation

An assumption based upon one population may not legitimately represent another.

Population limitations may concern: age, behavior, operating experience, physical characteristics, user group, system class, or another materially relevant population property.

Behavioral Limitation

Human, organizational, AI, autonomous-system, or environmental behavior may be represented through assumptions.

Where behavior changes beyond the assumed range, the assumption may cease to provide an adequate representation.

Operational Limitation

An assumption may apply only within a defined operating envelope.

For example:

Normal Operation

may not legitimately represent:

Emergency Operation

or:

Failure-State Operation.

Parameter Limitation

An assumption may depend upon parameter ranges.

Example:

Linear Response Assumption

may apply between:

Parameter X = 0–100

but not beyond that range.

SALM preserves this boundary explicitly.

Scenario Limitation

Some assumptions apply only to particular scenarios.

A scenario-specific assumption should not silently propagate into unrelated scenarios merely because it already exists in the simulation environment.

Evidence-Related Limitation

SABM may establish that an assumption depends upon limited evidence.

SALM governs the resulting applicability restriction where that evidence cannot legitimately support broader use.

Conceptually:

Limited Basis → Limited Applicability

Uncertainty-Derived Limitation

SAUM may establish uncertainty that materially constrains assumption use.

Where uncertainty becomes sufficiently significant, it may create: narrower applicability, stronger conditions, restricted interpretation, scenario restrictions, or a requirement for reassessment.

SAUM governs the uncertainty.

SALM governs the limitation that follows from it.

Dependency-Derived Limitation

SADM identifies what depends upon an assumption.

SALM determines where assumption limitations affect those dependencies.

Conceptually:

Assumption Limitation

↓

Dependency Path

↓

Affected Simulation Element

↓

Affected Scenario

↓

Affected Output / Conclusion Limitation Propagation

A limitation attached to an assumption may propagate through the simulation.

Example:

Battery Temperature Assumption Limited to −10°C to +45°C

↓

Battery Performance Model

↓

Power Availability

↓

Vehicle Range

↓

Mission Feasibility

A mission simulated outside the applicable temperature range cannot legitimately inherit the same assumption without reassessment.

No Limitation Disappearance Principle

SIMULOS® establishes:

A limitation attached to a material Simulation Assumption does not disappear when that assumption is embedded inside a model, scenario, computation, or simulation output.

The limitation must remain traceable wherever it materially constrains downstream interpretation.

Limitation Materiality

Not every limitation has equal significance.

Materiality may depend upon: affected assumption, dependency depth, dependency breadth, scenario significance, output sensitivity, consequence, intended simulation purpose, and Simulation Depth.

The governing question is:

Could crossing or ignoring this limitation materially change the simulation behavior, output, or legitimate interpretation?

Limitation Threshold

Where appropriate, SALM may establish a threshold at which a limitation becomes operationally significant.

Conceptually:

Within Boundary → Applicable

Near Boundary → Conditional / Monitor

Beyond Boundary → Reassessment / Non-Applicable

The exact states depend upon the simulation context.

Boundary Proximity

A simulation operating very close to an assumption boundary may require different treatment from one operating comfortably inside it.

SALM may therefore distinguish:

Inside Boundary

Boundary Proximity

Boundary Reached

Boundary Exceeded

This prevents binary treatment of limitations where proximity itself is materially relevant.

Limitation Breach

A Simulation Assumption Limitation Breach occurs when simulation conditions exceed an established applicability boundary.

A breach should trigger evaluation of: affected assumption, affected dependencies, affected scenarios, affected outputs, and required reassessment.

SALM does not automatically determine that the entire simulation is invalid.

It identifies the affected governance condition.

Limitation Interaction

Multiple assumption limitations may interact.

An individual assumption may remain within its own boundary while the combination of several limiting conditions creates a materially different simulation context.

Material interactions should therefore remain visible.

Compound Limitation

A Compound Limitation exists where multiple assumption limitations jointly constrain a simulation more strongly than each limitation considered independently.

This may be particularly important in: coupled systems, multi-agent simulations, environmental simulations, complex engineering simulations, and high-dimensional scenario spaces.

Limitation Concentration

A simulation may depend heavily upon a small number of assumptions with narrow applicability.

SALM may identify such concentration as a governance signal because small changes in context may cause large portions of the simulation to leave their supported assumption boundaries.

Output Limitation

An assumption limitation may constrain the meaning of simulation outputs.

For example:

Output applies only under Environment A

must not silently become:

Output applies universally.

SALM therefore preserves the relationship:

Assumption Boundary → Output Boundary

where materially applicable.

Interpretation Limitation

Simulation results may be technically produced outside an assumption's legitimate boundary.

Computational executability does not establish interpretive legitimacy.

SIMULOS® therefore establishes:

A simulation can produce an output under conditions where the assumptions supporting that output no longer justify its interpretation.

No Computational Legitimacy Principle

The ability of a simulation to execute does not prove that its assumptions remain applicable.

Execution capability and assumption applicability are separate governance conditions.

No Scope Inflation Principle

An assumption established for a bounded context must not be silently generalized beyond that context.

Conceptually:

Assumption Scope ≤ Governed Applicability Basis

Broader application requires an additional governed basis.

No Hidden Limitation Principle

A material assumption limitation must not be omitted merely because disclosure weakens the apparent usefulness, precision, coverage, or confidence of the simulation.

Limitation Disclosure

Material limitations should remain visible to downstream simulation governance.

Disclosure may include: limitation description, applicability boundary, limiting condition, affected assumption, affected dependencies, affected scenarios, affected outputs, materiality, boundary status, and required reassessment conditions.

Limitation Gap

A Simulation Assumption Limitation Gap exists where: applicability boundaries are unknown, material limiting conditions are unidentified, dependency effects are unclear, downstream consequences cannot be determined, or the assumption is being used outside its documented context without sufficient governance basis.

The gap should remain visible.

Unknown Boundary Principle

SIMULOS® establishes:

An unknown applicability boundary must not be represented as unlimited applicability.

Where the boundary cannot be established, the correct governance state may be:

Applicability Boundary Undetermined

rather than:

Universally Applicable.

Limitation Change

Assumption limitations may change when: the Simulation Object changes, the assumption changes, new evidence becomes available, uncertainty changes, dependencies change, model structure changes, simulation environment changes, scenarios change, operational conditions change, or new limiting conditions are discovered.

Limitation Reassessment

Material changes should trigger reassessment of affected assumption limitations.

The reassessment asks:

Does the existing applicability boundary still represent the conditions under which this assumption may legitimately operate?

Boundary Expansion

New evidence may support broader assumption applicability.

Conceptually:

Previous Boundary → New Basis → Expanded Boundary

Expansion should be governed rather than presumed.

Boundary Restriction

New evidence may demonstrate that an assumption has narrower applicability than previously believed.

Conceptually:

Previous Boundary → New Evidence → Restricted Boundary

Affected simulation outputs produced under the previous broader boundary may require reassessment.

Boundary Withdrawal

Where an assumption can no longer be supported within its previous context, its applicability boundary may be withdrawn.

The assumption may require: replacement, redesign, scenario restriction, or removal.

Version Binding

Material assumption limitations should remain bound to the relevant: Simulation Object version, assumption version, model version, simulation environment version, scenario version, and simulation configuration

where these distinctions affect applicability.

Historical Limitation Preservation

Later expansion or restriction of an assumption boundary should not erase the limitation state that applied when earlier simulations were executed.

Historical preservation supports: reconstruction, comparison, audit, incident analysis, and lifecycle governance.

Point-in-Time Limitation Reconstruction

SALM should support questions such as:

Which assumption boundary applied when this simulation was executed?

Was the simulated scenario inside or outside that boundary?

Had a limitation already been identified?

Was the assumption under reassessment?

Which outputs depended upon the affected assumption?

Limitation Traceability

SALM preserves the chain:

Simulation Assumption

↓

Applicability Boundary

↓

Limiting Condition

↓

Dependency

↓

Affected Simulation Element

↓

Scenario / Output

↓

Interpretation Limitation

where applicable.

Simulation Assumption Limitation Record

For each material limitation, SALM may establish: Limitation ID, Simulation Assumption ID, limitation description, limitation type, applicability boundary, applicable conditions, non-applicable conditions, limiting conditions, boundary threshold, boundary proximity, dependency references, affected model elements, affected scenarios, affected outputs, materiality, uncertainty reference, evidence-basis reference, limitation status, reassessment trigger, version binding, change history, and traceability references.

Simulation Assumption Limitation Map

Material limitations may be represented through a Simulation Assumption Limitation Map:

Assumptions → Applicability Boundaries → Dependencies → Scenarios → Outputs → Interpretation Boundaries

The map provides a structured view of where the simulation's assumption architecture ceases to support broader representation.

Limitation Status

A material limitation may receive a governed status such as:

Identified

Defined

Bounded

Conditional

Approaching Boundary

Boundary Reached

Boundary Exceeded

Under Reassessment

Restricted

Withdrawn

or:

Undetermined

according to the applicable governance context.

Relationship to SAIM

SAIM establishes:

Which Simulation Assumption exists?

SALM establishes:

Where does that assumption cease to remain applicable?

Relationship to SABM

SABM establishes:

Why is the assumption being used?

SALM establishes:

How far does that basis legitimately support its use?

Relationship to SADM

SADM establishes:

What depends upon the assumption?

SALM establishes:

Which dependencies become constrained when the assumption reaches its limitation boundary?

Relationship to SAUM

SAUM establishes:

What remains uncertain about the assumption?

SALM establishes:

Which applicability limitations follow from the assumption, including limitations created by material uncertainty.

Relationship to VALIDOS®

SALM does not determine whether the Simulation Assumption, simulation, output, or downstream claim satisfies general validation requirements.

VALIDOS® governs validation.

SALM provides the simulation-specific assumption limitation structure that validation may consume where relevant.

Relationship to ORA™

ORA™ governs actual Operational Reality.

SALM does not govern the real operational environment.

It governs whether a Simulation Assumption remains applicable to the represented conditions inside the simulation.

Relationship to INTEGROS®

INTEGROS® may govern integrity of limitation records, boundaries, states, and lifecycle continuity.

SALM governs the simulation-specific meaning and consequences of assumption limitations.

Minimum Implementation Framework

1. Identify Material Limitations

Determine which Simulation Assumptions contain limitations capable of materially affecting the simulation. 2. Establish Applicability Boundaries

Define where, when, and under which conditions each material assumption may legitimately operate. 3. Identify Limiting Conditions

Determine which changes or conditions narrow, terminate, or materially alter assumption applicability. 4. Connect Limitations to Dependencies

Use the governed dependency structure to identify affected model elements, scenarios, outputs, and conclusions. 5. Assess Limitation Materiality

Determine whether approaching, reaching, or exceeding the boundary can materially alter simulation behavior or interpretation. 6. Establish Boundary Status

Determine whether the assumption is inside, approaching, at, or beyond its governed applicability boundary. 7. Propagate Material Limitations

Ensure material limitations remain connected to affected downstream simulation elements and outputs. 8. Preserve Limitation Disclosure

Make material limitations visible wherever simulation interpretation materially depends upon them. 9. Reassess After Material Change

Reevaluate limitations when assumptions, evidence, uncertainty, dependencies, models, environments, or scenarios materially change. 10. Preserve Lifecycle Traceability

Maintain historical and current assumption-limitation records across simulation versions.

Governance Outputs

SALM may produce: Simulation Assumption Limitation Identifier, Simulation Assumption Limitation Record, Assumption Applicability Boundary, Applicability Condition Record, Non-Applicability Record, Limiting Condition Record, Environmental Limitation Record, Temporal Limitation Record, Spatial Limitation Record, Population Limitation Record, Behavioral Limitation Record, Operational Limitation Record, Parameter Limitation Record, Scenario Limitation Record, Evidence-Derived Limitation Record, Uncertainty-Derived Limitation Record, Dependency-Derived Limitation Record, Limitation Materiality Assessment, Limitation Threshold Record, Boundary Proximity Record, Limitation Breach Record, Limitation Propagation Record, Limitation Interaction Record, Compound Limitation Record, Output Limitation Record, Interpretation Limitation Record, Limitation Disclosure Record, Limitation Gap Record, Limitation Reassessment Record, Limitation Change Record, Simulation Assumption Limitation Map, Simulation Assumption Limitation Traceability Chain, and Governed Simulation Assumption Limitation Set.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

A pedestrian-behavior assumption is derived from observations of normal urban traffic conditions.

The simulation is later extended to emergency evacuation scenarios.
Application

SALM identifies that the assumption basis does not establish applicability under: panic behavior, mass pedestrian movement, emergency signaling, or abnormal traffic behavior.

The original assumption therefore has a bounded applicability context. Result

The simulation may continue to use the assumption for normal urban scenarios.

Its use in emergency evacuation scenarios becomes:

Applicability Boundary Exceeded — Reassessment Required

The simulation cannot silently generalize the original behavioral assumption into the new scenario class. Use Case 2 — Space Mission Simulation

Scenario

A spacecraft thermal model uses a component-performance assumption supported within a defined radiation and temperature envelope.

A new mission profile exposes the spacecraft to conditions outside that envelope. Application

SALM traces:

Component Assumption

↓

Thermal Boundary

↓

Power Subsystem

↓

Mission Scenario

↓

Mission Output

The original assumption remains technically usable by the simulation software, but its applicability boundary has been exceeded. Result

Affected mission outputs receive a governed limitation until the assumption is reassessed, replaced, or supported for the expanded environment.

Architectural Position

Simulation Assumption Limitation is the fifth and final internal governance space of the Simulation Assumption Standard (SAS).

The complete SAS progression is:

Simulation Assumption Identification → Simulation Assumption Basis → Simulation Assumption Dependency → Simulation Assumption Uncertainty → Simulation Assumption Limitation

Together, the five modules establish a complete assumption-governance chain:

What is assumed?

↓

Why is it assumed?

↓

What depends upon it?

↓

What remains uncertain?

↓

Where does its legitimate applicability end?

This progression ensures that Simulation Assumptions cannot function as invisible, unsupported, isolated, falsely certain, or unlimited representations of reality.

Foundational Principle

Every material Simulation Assumption has a boundary of legitimate applicability; when that boundary is unknown, approached, crossed, or changed, the limitation must remain visible wherever the simulation materially depends upon that assumption.

Canonical Closing Statement

Simulation Assumption Limitation Module completes the internal governance lifecycle of the Simulation Assumption Standard by establishing where Simulation Assumptions may legitimately operate, where their applicability ends, which conditions restrict their use, how those restrictions propagate through simulation dependencies, and how affected scenarios, outputs, and interpretations remain bounded by the assumptions upon which they depend. Through applicability-boundary governance, limiting-condition identification, materiality assessment, boundary proximity and breach detection, dependency propagation, limitation interaction, disclosure, reassessment, version binding, historical preservation, and complete limitation traceability, SALM prevents bounded Simulation Assumptions from being silently transformed into universal representations of reality and ensures that the limits of simulation assumptions remain governable throughout the complete simulation lifecycle.

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